

Density-Invariant Difference-DMD on Waddington-OT

Yifan Li

1 Dynamical Model

I used Waddington-OT's mean trajectories as data. Since the cell expressions were under the influence of OKSM, we can model it as:

$$x_{k+1} = Mx_k + b_k \quad (1)$$

where $x_k \in \mathbb{R}^G$ is a mean-centred trajectory, $M = (I + A)$ is the intrinsic unforced gene regulatory network operator, and b_k is the external control input (forcing) applied to the system at time step k .

2 Difference-DMD for State Isolation

The model assumes that external forcing b_k is uniformly applied to all parallel lineages at a given time point, so we can isolate the intrinsic dynamics:

$$x'_{k+1} - x_{k+1} = M(x'_k - x_k) \quad (2)$$

where x' and x are two parallel trajectories. Applying half-rank truncated Dynamic Mode Decomposition to these trajectory pair differences obtains a set of models $M_l = I + A_l$.

3 Geometric Fusion

We then obtain the global model via fusing these matrices:

$$A_{fused} = \left(\sum_l A_l \right) \left(\sum_l P_l + \epsilon I \right)^{-1} \quad (3)$$

This fused matrix is the optimal solution to:

$$\min_A \sum_l \|(A - A_l)P_l\|_F^2 \quad (4)$$

where P_l is the orthogonal projector onto the lineage's subspace. Compared to the usual practice of stacking parallel lineages then fitting a single DMD matrix, this method prevent prevalent stagnant lineages (e.g., stagnant stroma) from mathematically diluting rare successful trajectories, by only considering A_l in the subspace it is informed on.

4 Validation and Physical Constraints

4.1 Multi-Horizon Predictive Validity

To evaluate structural limits, models were tested out-of-sample (20% lineage holdout) across $h \in [1, 36]$ steps. I tested the method along with 4 other simplified model: the same model, but with a constant forcing term fitted; the same model, without the forcing term, so just DMD; just a constant forcing term; just a time varying forcing term. As shown in Table 1, the original model has $r^2 > 0.6$ consistently; none of the simplified model managed to stay close to data as rolled forward, all of them diverged.

Table 1: Out-of-Sample Multi-Horizon Comparison (20% Holdout). Evaluating predictions across h steps for cross-lineage variance (R^2). The time-varying model ($M + b_t$) successfully prevents compounding errors at long horizons.

Model	$h = 1$	$h = 2$	$h = 4$	$h = 8$	$h = 16$	$h = 32$	$h = 36$
Panel A: $R^2(h)$ (Cross-Lineage Variance)							
$M + b_t$ (Time-Varying)	+0.714	+0.798	+0.694	+0.617	+0.445	+0.602	+0.636
Pure DMD (No Control)	+0.633	+0.416	-1.177	-20.488	-144.351	-13.955	-20.495
$M + B_{const}$ (LTI)	-2.639	-14.741	-40.941	-338.500	-685.676	-93.141	-74.299
Pure Forcing b_t	-8.163	-32.897	-91.199	-836.520	-1876.644	-586.861	-469.015
Pure Forcing B_{const}	-10.935	-54.764	-165.641	-1695.702	-3196.395	-632.030	-486.110

4.2 Dimensionality of Control

Beyond predictiveness on lineage differences, by obtaining $M_{fused} = I + A_{fused}$ we can also recover the forcing term b_k :

$$b_k = x_{k+1} - M_{fused}x_k \quad (5)$$

In linear control theory, $b_k = Bu_k$, where u_k represents Transcription Factors and the matrix B translates TF binding into genome-wide expression influences. From this model, one expects the variance of b_k over time to be roughly 4 dimensional or less; PCA on b_k shows that the top 4 PC explains 75% of variance.

4.3 Temporal Structure and Biological Validation of the Control Input

To rigorously verify that the mathematically extracted control input (b_k) represents true biological forcing rather than an artifact of the decomposition, I established strict, pre-registered temporal hypotheses prior to projecting the data. These hypotheses posit the peak timing of canonical markers across distinct functional classes: exogenous OSKM delivery (*Pou5f1*, *Sox2*, *Klf4*, *Myc*), mesenchymal suppression (e.g., *Col1a1*, *Fn1*), and epithelial activation (e.g., *Krt8*, *Cdh1*).

- The exogenous OSKM signature strictly peaks at $t = 0.0$, as predicted by the early viral payload onset.
- The mesenchymal expression nadir occurs on Day 3.0, preceding the epithelial activation peak on Day 8.5, accurately capturing the sequence of the Mesenchymal-to-Epithelial Transition.

One can also track the geometric alignment and magnitude of the forcing over time (Figure 1). As predicted by the biological sequence, the absolute magnitude of the control energy ($\|b_k\|_2$)

is maximized immediately during the initial viral payload delivery, which dominates the temporal average B_{const} . The angular disalignment in later stages is also predicted, as late-stage reprogramming orthogonalizes away from the primary MET axis, which is confirmed by the collapsing cosine similarity of B_{const} with b_k .

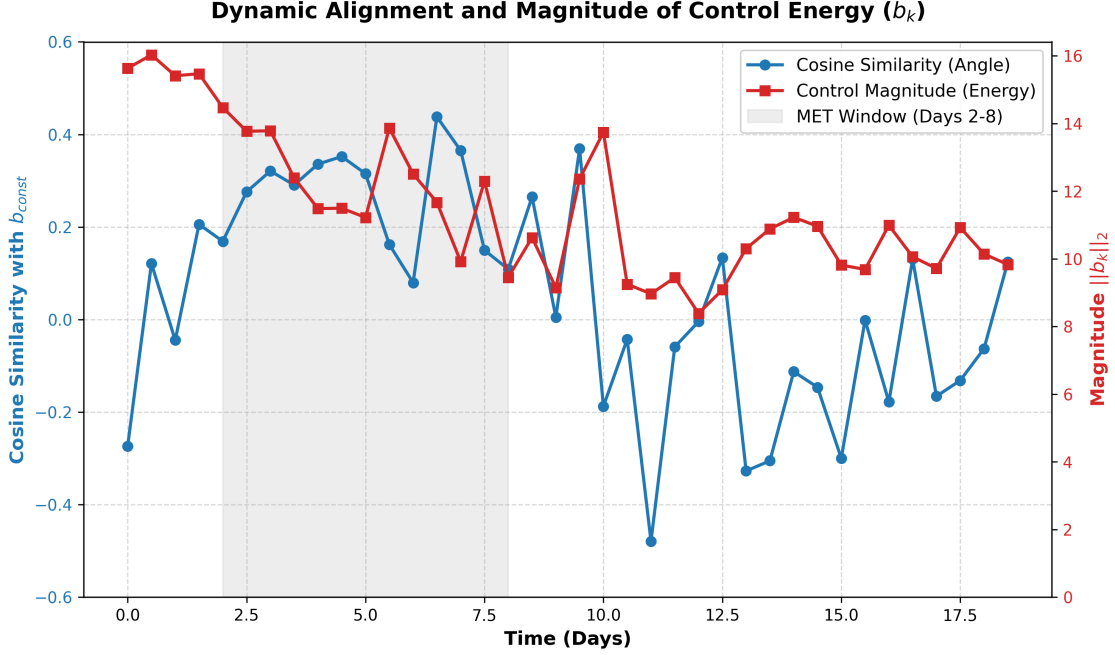


Figure 1: Dynamic alignment and magnitude of step-by-step forcing (b_k) relative to the global average (B_{const}). The absolute control energy (magnitude) peaks immediately at $t = 0$, while the geometric direction (cosine similarity) strictly surges within the established biological MET window (Days 2-8), demonstrating the physical time-boundaries of the primary structural drive.